

APM Project Management Qualification

Learner Study Pack



Solutions Development

Introduction

This section of the Learner Study Pack will cover the Solutions Development competence of the APM Project Management Qualification.

This section will enable you to gain an understanding of the process of clarifying the problem, exploring options to confirm a preferred solution.

Learning objective and outcomes

Learning objective:

17. Understand solutions development as the ability to determine the optimal solution to satisfy agreed requirements.

By the end of this activity, you should be able to:

17a) Understand how to evaluate and prioritise requirements in order to deliver the optimal solution

And

Understand the different approaches for different life cycle models, e.g. the use of MVP and MMP in iterative life cycles.

Study time

This section of the Learner Study Pack will take approximately 1 - 2 hours to complete.

Recommended resources

In addition to your Learner Study Pack, use the below references for more information on Solutions Development:

- APM Body of Knowledge Seventh edition, 4.1.3 (Options and solutions)
- APM Learning Module, Developing effective approaches and solutions. Accessed at: <https://www.apm.org.uk/resources/apm-learning/>
- APM Project Journal, Time for Speed. Accessed at: <https://www.apm.org.uk/news/time-for-speed/>
- Edward de Bono (2009) *Lateral Thinking: A Textbook of Creativity*.
- UK Government (2022) *The Green Book* [gov.uk/government/publications/the-green-book-appraisal-and-evaluation-in-central-government](https://www.gov.uk/government/publications/the-green-book-appraisal-and-evaluation-in-central-government)

Overview

Solutions development in project management is a systematic process aimed at identifying, analysing and implementing effective responses to challenges and opportunities within a project.

This approach involves innovative problem-solving, risk mitigation and the creation of strategies.

By encouraging adaptability and addressing issues pro-actively, solutions development helps with the success and overall resilience of the project.

Evaluate and Prioritise to Deliver Solution

The Business case documents the options considered including an option to 'do nothing'.

Solutions development requires clarity on the problem to be solved and then exploring different options until a preferred solution is identified and subsequently maintained and refined

When evaluating and prioritising you should consider the following:

- Priority of objectives and requirements.
- Acceptance criteria necessary to be achieved.
- Priorities of the end users
- The life cycle approach for project delivery.
- Value of alternative solutions.
- Potential for stakeholder pressure to adopt a 'preferred solution' which may not be the optimal choice.

Defining Different Options

Each project presents unique challenges, requirements, and constraints, requiring careful consideration of various approaches and alternatives.

By exploring different options, project managers can thoroughly evaluate potential strategies, methodologies, technologies, and resources available for achieving project outcomes.

This not only enhances understanding of the project but also allows for informed decision-making based on factors such as cost-effectiveness, feasibility, scalability, and risk mitigation.

When defining the different options, there may also be different criteria to consider as well. Typically, these are dependent on the type of project, but may include:

- Technical - Process, material, software, future.
- Social - People, interactions, needs, relationships.
- Procurement - Lease, buy, make, service, partnership.
- Management - Life cycle, governance.
- Transition - Phased, all at once 'big bang.'

Prioritising A

If the project is unable to deliver all the requirements within its scope, the requirements may have to be prioritised.

A common technique to do this is the **MoSCoW** approach:

- **M** - Must have
- **S** - Should have
- **C** - Could have
- **W** - Won't have

This process is designed to ensure that essential requirements are understood as an input to the work to allow for the selection of the optimal solution.

Once options have been identified, they can then be prioritised by considering:

- An investment appraisal – considering the trade-off between costs, benefits and risks, looking at value management techniques focusing on the value that requirements contribute to the overall objective.
- Stakeholder commitment – to the solution.
- Market capability – considering supply of materials and/or services required for the project

Other prioritisation techniques that can be used to establish importance are:

- Numerical assignment – grouping requirements.
- The ABC model – classifying objects based on importance.
- The 5 Whys method – asking the stakeholders 'why the requirement is necessary'.

Different Approaches for Different Life cycles

Projects using different life cycles take different approaches to developing solutions:

In a linear life cycle, analysis of options is done early on for the selection of a preferred optimum solution to be agreed with scope. This is then confirmed, and any changes thereafter would be managed through change control.

For an iterative life cycle, there is the opportunity to explore options for longer, allowing different design options to be investigated while the needs are being investigated in another timebox. Iterative development provides an environment whereby the solution can be delivered in increments providing opportunity for 'Must have' requirements to be delivered as early as possible, in the form of a minimum viable product (MVP).

The project can then gain progressive user feedback and the gradual delivery of requirements in the next steps of product development to achieve a minimum marketable product (MMP) through to future versions of the product eventually delivering all requirements.

Scenario One:

The work onsite for the 'Snappy Appy' renovation project is about to begin, but as the building is Grade II listed, there is the anticipation that there may be potential problems.

The main builder has just been to see you and outlined the regulatory requirements that the project must follow.

You already have a set of requirements from the organisation, some of which conflict with each other, and with the regulatory requirements.

Therefore, you need to balance all of these requirements to deliver the best solution.

What factors would you need to consider when evaluating the requirements and defining different options for the solution?

Scenario Two:

Ayo, the Founder, has approached you with a list of requirements that they would like included in the renovation project.

They want the new head office of 'Snappy Appy' to be perfect and so want all of them included.

They would like the following included:

- A quiet space that can be used by employees to practice mindfulness throughout the day.
- Interactive whiteboards in all the meeting rooms to allow for greater collaboration and creativity.
- A green wall in the communal kitchen to promote wellbeing and calm.
- Sitting/standing adjustable desks for the Customer Services Team in line with Occupational Health and Safety guidelines.

What techniques could you use to prioritise Ayo's requirements?

Using one of these techniques, how would you prioritise the various requirements?

Learning review questions

Before you attempt the exam practice questions in the next section, we recommend you review your learning to quickly see which aspects you don't understand or can't remember as well.

It will also boost your confidence by confirming what you have learnt and your demonstrated ability to apply that knowledge, so that you are ready to practise for the exam.

Here are some questions you may wish to answer to review your learning from the Solutions Development section:

- What are the different prioritisation and evaluation techniques you can use to confirm an optimal solution?
- How might an iterative life cycle influence a potential solution?
- What different criteria could be used to distinguish priorities when considering options ?

Exam practice questions

17.1 You have started a new role as Project Manager, how would you evaluate and prioritise your new project's objectives and requirements in order to deliver the optimal solution? (1 Mark)

- a. I will prioritise objectives and requirements based on their ease of implementation. This will help us get quick wins.
- b. I will prioritise based on historical data from similar projects. What worked in the past should work here too.
- c. I will check requirements are "SMART" and prioritise into Must haves, Should haves, Could haves and Won't haves.
- d. All objectives and requirements are important, so I will prioritise them equally to ensure a balanced approach.

17.2 You are managing a project to launch a new mobile application for a fast-food company. A rival has already launched its own mobile application, receiving great feedback from customers. Because of this, there is a lot of pressure to develop your solution as quickly as possible.

Having gathered the initial project requirements, the project team prioritises the most important solution items to build first based on the ____ (a) ____.

With the requirements prioritised, you decide to adopt an iterative approach, with a target to launch the first version of the solution in just four weeks. This will enable the project team to ____ (b) ____.

(2 marks)

(a)

- 1. Appetite for risk
- 2. Rival application
- 3. Quality management plan
- 4. Value to the customer

(b)

- 1. Gain feedback to inform the next release
- 2. Appease their stakeholders
- 3. Save resource by building part of solution
- 4. Reduce the overall project scope

Exam practice answers

Question	Answer / Mark scheme
17.1	C
17.2	(a) 4 (b) 1